Asset Pricing Theory

MFE5340 - AI in FE: Quantitative Investment

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Contents

1 Asset Returns **2**

2 Mean-Variance Analysis and CAPM **2**
2.1 Mean-Variance Analysis 2
2.2 Mean-Variance Frontier and Single Factor Model 7
2.3 Capital Asset Pricing Model 8

3 Stochastic Discount Factor **9**
3.1 Concept 9
3.2 Mean-Variance Frontier and SDF 10

4 Multi-Factor Models **11**

5 Ad-Hoc Factor Models **14**

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1 Asset Returns

Let P_{it} denote the price of some asset i (e.g., a stock) at time t. Let D_{it} be its dividend payment between $t-1$ and t. Then, the **net** return of this asset at t is:

$$R_{it} = \frac{P_{it} + D_{it} - P_{it-1}}{P_{it-1}} = \frac{P_{it} + D_{it}}{P_{it-1}} - 1. \tag{1}$$

Please be careful that here I use R to denote the net return of an asset, while in some (and maybe many) other materials, R is used to denote the **gross** return, which is $1 +$ net return. We usually use R_f to denote the risk-free rate (i.e., the net return of the risk-free asset) and assume it is constant.

Let $p_{it} = \log P_{it}$. Then, the *logarithmic* return or continuously compounded return on an asset is:

$$r_{it} = \log(P_{it} + D_{it}) - \log(P_{it-1}). \tag{2}$$

It is easy to show that the relationship between R_{it} and r_{it} is $r_{it} = \log(1 + R_{it})$ or $R_{it} = \exp(r_{it}) - 1$. When your time inverval from $t-1$ to t is small (e.g., you are calculating returns at daily frequency or higher frequencies), then R_{it} and r_{it} are very close to each other.

Another related concept is **excess** return. If you borrow a dollar at the risk-free rate R_f and invest it in an asset with return R_i, you pay no money out-of-pocket today, and get the payoff $R_{it} - R_f$. In asset pricing, factors are constructed as long/short portfoilos of assets (i.e., stocks). This long/short portfolio also generates an excess return. We often use the notation R^e_{it} for excess returns. Asset pricing is about how we can find a model for the expected excess return, i.e., $\mathbb{E}[R^e_{it}]$.

2 Mean-Variance Analysis and CAPM

Much empirical work in asset pricing is written in terms of mean-variance frontiers and beta factor model representations (Cochrane 2005). We tackle these concepts in this and following sections. First, let's look at mean-variance analysis, which derives the mean-variance frontier. Additionally, within the framework of mean-variance analysis, we will also introduce one of the most important models in asset pricing: the Capital Asset Pricing Model (CAPM).

2.1 Mean-Variance Analysis

Mean-variance analysis, introduced by Markowitz (1952), is a cornerstone of modern portfolio theory. It provides a framework for constructing portfolios that optimize expected return (means) for a given level of risk (measured by standard deviation/variance of return), or equivalently, minimize risk for a given level of expected return. It generates a mean-variance frontier which is the boundary of the set of means and variances of the returns on all portfolios of the given assets. Many asset pricing propositions and test statistics have interpretations in terms of the mean-variance frontier.

2.1.1 Mean-Variance Optimization

We start with a vector of N asset returns $\boldsymbol{R}$. (We suppress the time index t as we consider a fixed time here.) Let $\boldsymbol{\mu}$ represent the expected returns of the assets, i.e., $\boldsymbol{\mu} = \mathbb{E}[\boldsymbol{R}]$ and $\boldsymbol{\Sigma}$ be the covariance matrix of $\boldsymbol{R}$. Then, a portfolio of these assets is defined by its weights $\boldsymbol{w}$. The portfolio's return is $\boldsymbol{w}^\top \boldsymbol{R}$, and the portfolio's variance is $\boldsymbol{w}^\top \boldsymbol{\Sigma} \boldsymbol{w}$. With these notations, the mean-variance analysis can be stated as "*choose a portfolio to minimize variance for a given mean*":

$$\begin{array}{rl} \min\limits_{\boldsymbol{w}} & \boldsymbol{w}^\top \boldsymbol{\Sigma} \boldsymbol{w} \\ \text{subject to} & \boldsymbol{w}^\top \boldsymbol{\mu} = \mu^\star \\ & \boldsymbol{w}^\top \mathbf{1} = 1, \end{array} \tag{3}$$

where $\mu^\star$ is the target expected return of the portfolio. We can solve this problem by Lagrangian multipliers. The Lagrange function ($\mathcal{L}$) for this optimization problem is:

$$\boldsymbol{w}^\top \boldsymbol{\Sigma} \boldsymbol{w} - 2\lambda(\boldsymbol{w}^\top \boldsymbol{\mu} - \mu^\star) - 2\gamma(\boldsymbol{w}^\top \mathbf{1} - 1). \tag{4}$$

Differentiating this objective function with respect to $\boldsymbol{w}$ and setting the derivative to 0 leads to:

$$\boldsymbol{\Sigma} \boldsymbol{w} - \lambda \boldsymbol{\mu} - \gamma \mathbf{1} = 0, \tag{5}$$

or

$$\boldsymbol{w} = \boldsymbol{\Sigma}^{-1}(\lambda \boldsymbol{\mu} + \gamma \mathbf{1}). \tag{6}$$

We can find the Lagrangian multipliers by plugging $\boldsymbol{w}$ into the two constraints:

$$\begin{array}{rcl} \boldsymbol{\mu}^\top \boldsymbol{\Sigma}^{-1}(\lambda \boldsymbol{\mu} + \gamma \mathbf{1}) & = & \mu^\star \\ \mathbf{1}^\top \boldsymbol{\Sigma}^{-1}(\lambda \boldsymbol{\mu} + \gamma \mathbf{1}) & = & 1. \end{array} \tag{7}$$

Since we have two equations and two unknowns, they can be solved. Define $A = \boldsymbol{\mu}^\top \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}$, $B = \boldsymbol{\mu}^\top \boldsymbol{\Sigma}^{-1} \mathbf{1}$ and $C = \mathbf{1}^\top \boldsymbol{\Sigma}^{-1} \mathbf{1}$, then the solutions for λ and γ are:

$$\begin{array}{rcl} \lambda & = & \dfrac{C\mu^\star - B}{AC - B^2} \\ \gamma & = & \dfrac{A - B\mu^\star}{AC - B^2}. \end{array} \tag{8}$$

Plugging them back to the expression for $\boldsymbol{w}$:

$$\begin{array}{rcl} \boldsymbol{w}^\star & = & \boldsymbol{\Sigma}^{-1}(\lambda \boldsymbol{\mu} + \gamma \mathbf{1}) \\ & = & \dfrac{C\mu^\star - B}{AC - B^2} \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu} + \dfrac{A - B\mu^\star}{AC - B^2} \boldsymbol{\Sigma}^{-1} \mathbf{1}. \end{array} \tag{9}$$

We can do some trick and re-write the expression as:

$$\begin{aligned} \boldsymbol{w}^\star &= \frac{C\mu^\star - B}{AC - B^2} B \frac{\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}}{B} + \frac{A - B\mu^\star}{AC - B^2} C \frac{\boldsymbol{\Sigma}^{-1}\mathbf{1}}{C} \\ &= \left(\frac{BC\mu^\star - B^2}{AC - B^2}\right) \frac{\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}}{\boldsymbol{\mu}^\top \boldsymbol{\Sigma}^{-1}\mathbf{1}} + \left(\frac{AC - BC\mu^\star}{AC - B^2}\right) \frac{\boldsymbol{\Sigma}^{-1}\mathbf{1}}{\mathbf{1}^\top \boldsymbol{\Sigma}^{-1}\mathbf{1}}. \end{aligned} \tag{10}$$

This way, the optimal weights $\boldsymbol{w}^\star$ can be viewed as the weighted average of two special portfolios $\boldsymbol{w}_1 = \frac{\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}}{\boldsymbol{\mu}^\top \boldsymbol{\Sigma}^{-1}\mathbf{1}}$ and $\boldsymbol{w}_2 = \frac{\boldsymbol{\Sigma}^{-1}\mathbf{1}}{\mathbf{1}^\top \boldsymbol{\Sigma}^{-1}\mathbf{1}}$ (note that the denominators in these two expressions are just the normalization constant). Besides, the coefficients of these two portfolios sum to one, i.e., $\frac{BC\mu^\star - B^2}{AC - B^2} + \frac{AC - BC\mu^\star}{AC - B^2} = 1$. As a result, the optimal portfolio $\boldsymbol{w}^\star$ is a combination of $\boldsymbol{w}_1$ and $\boldsymbol{w}_2$, and the weights are uniquely determined by $\mu^\star$. This phenomenon is also known as the **two-fund separation theorem**.

The two-fund separation theorem highlights the efficiency and simplicity of the mean-variance optimization framework. It underscores the idea that, regardless of individual risk preferences (i.e., different choices of $\mu^\star$), all investors will hold a combination of the same two basic portfolios. Certainly, the basis portfolios are not unique, meaning we do not have to choose $\boldsymbol{w}_1$ and $\boldsymbol{w}_2$ in the manner described above. However, the $\boldsymbol{w}_1$ and $\boldsymbol{w}_2$ mentioned above have some special properties. For example, $\boldsymbol{w}_2$ is the global minimum variance portfolio, and $\boldsymbol{w}_1$ gives you the highest risk-adjusted expected return that you can achieve by investing in all risky assets.

Apparently, we can solve for different $\boldsymbol{w}^\star$ for different $\mu^\star$, and for each $\boldsymbol{w}^\star$ we can calculate its standard deviation (which is the square root of variance), i.e., $\sigma_p = \sqrt{(\boldsymbol{w}^\star)^\top \boldsymbol{\Sigma} \boldsymbol{w}^\star}$. The pairs (σ_p, μ_p), where $\mu_p = \mu^\star$, form the minimum-variance frontier of the **risky** assets, with the upper half of this hyperbolic curve being the mean-variance efficient frontier in the absense of a risk-free asset. Figure 1 demonstrates the mean-variance frontier constructed by the risky assets.

2.1.2 Adding the Risk-Free Rate

Suppose there is a risk-free asset R_f and further assume that it is a constant. In this case, the optimal portfolio is a combination between all risky assets and the risk-free asset. We still let $\boldsymbol{w}$ represent the weights of the risky assets, and then the weight on the risk-free asset is $1 - \boldsymbol{w}^\top \mathbf{1}$. This expression has two implications: (1) the weights of all assets including both risky and risk-free assets sum to 1; (2) it can be negative, which means that we are allowed to borrow money at the risk-free rate to invest in the risky assets. Now, the expected value of this portfolio is $\boldsymbol{w}^\top \boldsymbol{\mu} + (1 - \boldsymbol{w}^\top \mathbf{1}) R_f$ and the objective function becomes:

$$\begin{aligned} \min_{\boldsymbol{w}} \quad & \boldsymbol{w}^\top \boldsymbol{\Sigma} \boldsymbol{w} \\ \text{subject to} \quad & \boldsymbol{w}^\top \boldsymbol{\mu} + (1 - \boldsymbol{w}^\top \mathbf{1}) R_f = \mu^\star. \end{aligned} \tag{11}$$

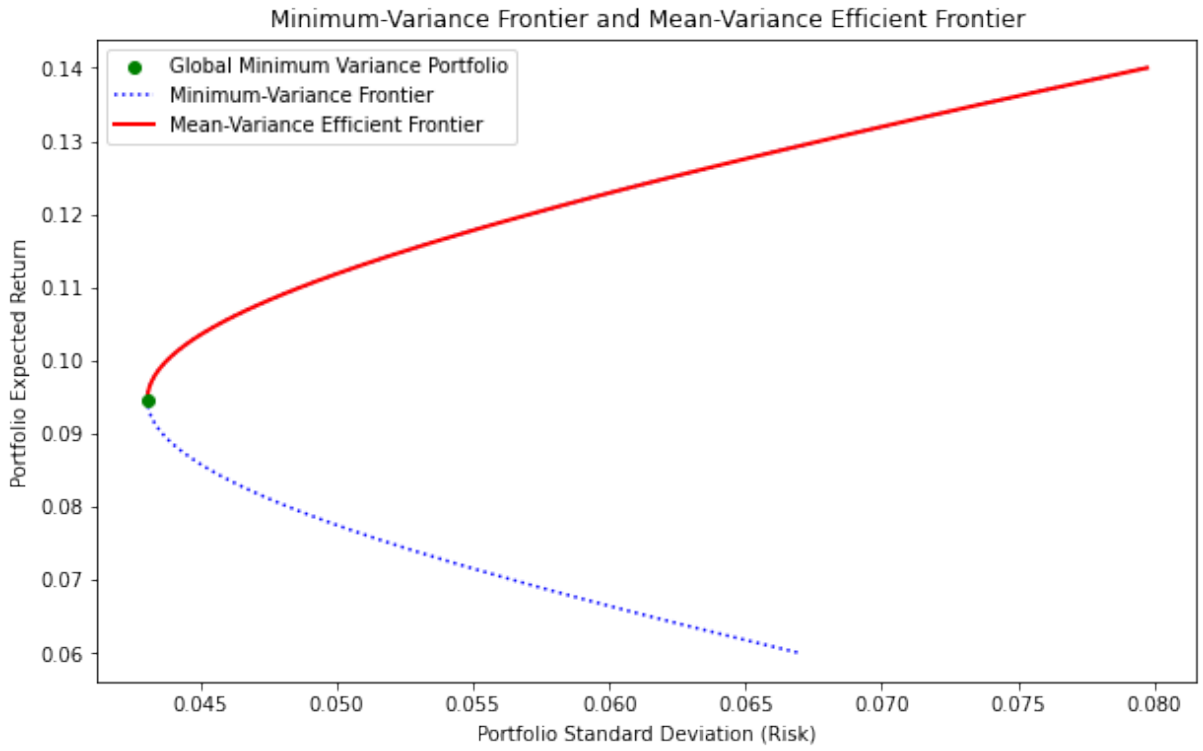


Figure 1: Mean-Variance Frontier

Or we can rearrange the constraint and use $\boldsymbol{\mu}^e = \boldsymbol{\mu} - R_f\mathbf{1}$ to denote the expected excess return:

$$\begin{aligned} \min_{\boldsymbol{w}} \quad & \boldsymbol{w}^\top \boldsymbol{\Sigma} \boldsymbol{w} \\ \text{subject to} \quad & \boldsymbol{w}^\top \boldsymbol{\mu}^e = \mu^\star - R_f. \end{aligned} \tag{12}$$

Note that we no longer need the constraint $\mathbf{w}^\top\mathbf{1} = 1$. (Why?)

The optimal weights are (the derivation is left to you as an assignment):

$$\boldsymbol{w}^\star = \left(\frac{\mu^\star - R_f}{(\boldsymbol{\mu}^e)^\top \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}^e} \right) \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}^e. \tag{13}$$

We can do some trick again to express $\boldsymbol{w}^\star$ as a scaled version of a special portfolio:

$$\mathbf{w}^\star = \left(\frac{(\mu^\star - R_f)\mathbf{1}^\top \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}^e}{(\boldsymbol{\mu}^e)^\top \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}^e} \right) \frac{\boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}^e}{\mathbf{1}^\top \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}^e}. \tag{14}$$

Let $\boldsymbol{w}_{\tan} = \frac{\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}^e}{\mathbf{1}^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}^e}$ denote this special portfolio (again the denominator is just the normalization constant). Then $\boldsymbol{w}^\star = \delta \boldsymbol{w}_{\tan}$ where $\delta = \frac{(\mu^\star - R_f)\mathbf{1}^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}^e}{(\boldsymbol{\mu}^e)^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}^e}$ is a scaling factor. Since the sum of the weights in $\boldsymbol{w}_{\tan}$ is one, it is easy to see that the sum of the weights in $\boldsymbol{w}^\star$, i.e., $(\boldsymbol{w}^\star)^\top\mathbf{1}$, equals δ.

So, is there any intuition behind the portfolio $\boldsymbol{w}_{\tan}$? The answer is a big yes. If we plot the portfolio's standard deviation σ on the horizontal axis and the expected return on the vertical axis, and draw a tangent line from the risk-free rate (a point located at $(0, R_f)$) to the mean-variance frontier formed by risky assets, the coordinates of the tangency point are exactly the risk and return

associated with $\boldsymbol{w}_{\tan}$. Therefore, this portfolio is also known as the **tangency portfolio**.

Reflecting on our initial problem, we aim to construct a combination of risky assets and the risk-free asset given a target $\mu^\star$, with weights $\boldsymbol{w}^\star$ and $1-\delta$ respectively. Using the tangency portfolio mentioned above, **our portfolio effectively allocates $1-\delta$ of the funds to the risk-free asset (borrowing if the weight is negative) and δ of the funds to $\boldsymbol{w}_{\tan}$**. In other words, any portfolio with a given expected return is a combination of the risk-free asset and the tangency portfolio. This is the essence of the two-fund separation theorem in the presence of a risk-free asset (or one-fund theorem since the other one is the risk-free asset). Figure 2 shows the mean-variance frontier with a risk-free asset.

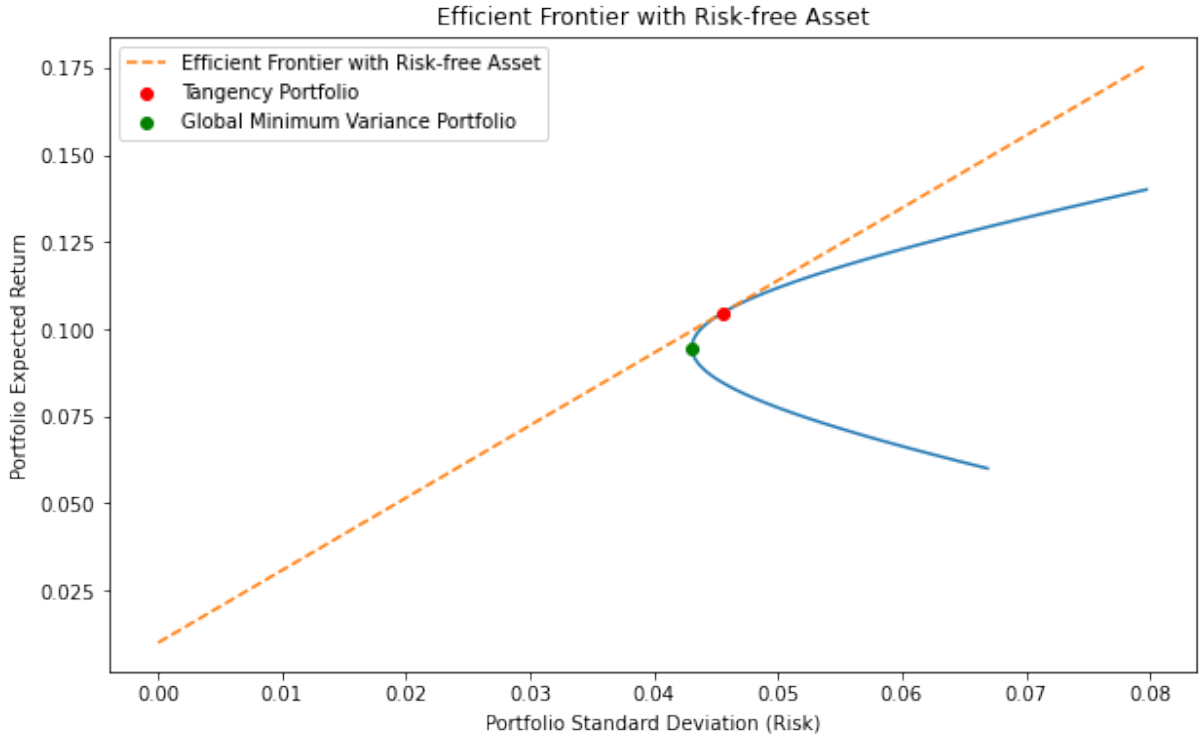


Figure 2: Mean-Variance Frontier with a Risk-free Asset

In finance, for an asset (either an individual security or a portfolio) with an expected return of μ and variance σ^2, the Sharpe ratio is defined as

$$\text{Sharpe Ratio} = \frac{\mu - R_f}{\sigma}. \tag{15}$$

We can see that the **slope** of the tangent line that connects the risk-free asset and the tangency portfolio indeed represents the highest Sharpe ratio that can be achieved by these risky assets and the risk-free asset. This straight line is the efficient frontier when there is a risk-free asset. It is also called the capital market line (CML). Let M denote the tangency point, then CML is defined by the following linear equation:

$$\mu = R_f + \frac{\mu_M - R_f}{\sigma_M}\sigma. \tag{16}$$

2.2 Mean-Variance Frontier and Single Factor Model

From the mean-variance frontier, we can derive a single factor model that can be used to price individual assets. This is a very important result in asset pricing theory. As a matter of fact, for any portfolio on the mean-variance frontier (other than the risk-free asset, and not necessarily the tangency portfolio), we can construct a single factor model (Cochrane 2005):

$$\mu_i - R_f = \beta_i(\mathbb{E}[R_{\text{mv}}] - R_f), \tag{17}$$

where R_{mv} represents the return of some portfolio on the mean-variance frontier. We can derive this result from different approaches (see Cochrane 2005), but I pick the following approach for its elegance and simplicity (I'd like to give full credit to Prof. Wang, Xi of Peking University, School of Economics, for this derivation).

Consider a portfolio $\boldsymbol{w}_{\text{mv}}$ on the mean-variance frontier. Now, we know that this portfolio can be viewed as a combination of the risk-free asset and the tangency portfolio, i.e., $\boldsymbol{w}_{\text{mv}} = [1-\delta, \delta\boldsymbol{w}_{\tan}^\top]^\top$, where $\delta\boldsymbol{w}_{\tan}$ represents the weights of the risky assets in $\boldsymbol{w}_{\text{mv}}$. With the expression of $\boldsymbol{w}_{\tan}$, $\delta\boldsymbol{w}_{\tan}$ can be written as:

$$\delta\boldsymbol{w}_{\tan} = \delta\frac{\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}^e}{\mathbf{1}^\top\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}^e} = \delta X\boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu} - R_f\mathbf{1}) = \delta X\left(\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu} - R_f\boldsymbol{\Sigma}^{-1}\mathbf{1}\right), \tag{18}$$

where $X = (\mathbf{1}^\top\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}^e)^{-1}$.

Now, for any asset i: $R_i = \boldsymbol{e}_i^\top\boldsymbol{R}$, where $\boldsymbol{e}_i$ is a vector with a 1 in the ith position and 0 elsewhere, its covariance with R_{mv} is (note that risk-free asset in this mean-variance portfolio is a constant):

$$\begin{aligned}
\text{cov}(R_i, R_{\text{mv}}) &= \text{cov}(\boldsymbol{e}_i^\top\boldsymbol{R}, \delta\boldsymbol{w}_{\tan}^\top\boldsymbol{R}) \\
&= \boldsymbol{e}_i^\top\boldsymbol{\Sigma}\delta\boldsymbol{w}_{\tan} \\
&= \boldsymbol{e}_i^\top\boldsymbol{\Sigma}\delta X\left(\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu} - R_f\boldsymbol{\Sigma}^{-1}\mathbf{1}\right) \\
&= \delta X\boldsymbol{e}_i^\top\boldsymbol{\Sigma}\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu} - \delta X R_f\boldsymbol{e}_i^\top\boldsymbol{\Sigma}\boldsymbol{\Sigma}^{-1}\mathbf{1} \\
&= \delta X\underbrace{\boldsymbol{e}_i^\top\boldsymbol{\mu}}_{\mathbb{E}[R_i]} - \delta X R_f\underbrace{\boldsymbol{e}_i^\top\mathbf{1}}_{1} \\
&= \delta X(\mathbb{E}[R_i] - R_f).
\end{aligned} \tag{19}$$

Next, let's look at the variance of R_{mv}:

$$\begin{aligned}
\text{var}(R_{\text{mv}}) &= \text{cov}(\delta\boldsymbol{w}_{\tan}^\top\mathbf{R}, \delta\boldsymbol{w}_{\tan}^\top\boldsymbol{R}) \\
&= \delta\boldsymbol{w}_{\tan}^\top\boldsymbol{\Sigma}\delta\boldsymbol{w}_{\tan} \\
&= \delta^2 X\underbrace{X\left(\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu} - R_f\boldsymbol{\Sigma}^{-1}\mathbf{1}\right)^\top}_{\boldsymbol{w}_{\tan}^\top}\underbrace{\boldsymbol{\Sigma}\left(\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu} - R_f\boldsymbol{\Sigma}^{-1}\mathbf{1}\right)}_{\boldsymbol{\mu} - R_f\mathbf{1}} \\
&= \delta X\delta\underbrace{\boldsymbol{w}_{\tan}^\top(\boldsymbol{\mu} - R_f\mathbf{1})}_{\mathbb{E}[R_{\tan}] - R_f} \\
&= \delta X\delta(\mathbb{E}[R_{\tan}] - R_f).
\end{aligned} \tag{20}$$

Further, we notice that $\mathbb{E}[R_{\text{mv}}] - R_f = (1-\delta)R_f + \delta\mathbb{E}[R_{\tan}] - R_f = \delta(\mathbb{E}[R_{\tan}] - R_f)$. Plugging this into the expression of $\text{var}(R_{\text{mv}})$ gives:

$$\text{var}(R_{\text{mv}}) = \delta X(\mathbb{E}[R_{\text{mv}}] - R_f). \tag{21}$$

Now, we have both ingredients to derive the single factor model:

$$\begin{aligned} \text{cov}(R_i, R_{\text{mv}}) &= \delta X(\mathbb{E}[R_i] - R_f), \\ \text{var}(R_{\text{mv}}) &= \delta X(\mathbb{E}[R_{\text{mv}}] - R_f). \end{aligned} \tag{22}$$

Dividing the two equations yields:

$$\beta_i \equiv \frac{\text{cov}(R_i, R_{\text{mv}})}{\text{var}(R_{\text{mv}})} = \frac{\mathbb{E}[R_i] - R_f}{\mathbb{E}[R_{\text{mv}}] - R_f}, \tag{23}$$

or, finally,

$$\mathbb{E}[R_i] - R_f = \beta_i(\mathbb{E}[R_{\text{mv}}] - R_f). \tag{24}$$

This is the single factor model associated with the mean-variance portfolio R_{mv}.

2.3 Capital Asset Pricing Model

Now, we are ready to introduce the Capital Asset Pricing Model (CAPM). CAPM has the same expression as the above equation, except that the mean-variance portfolio is replaced by the market portfolio:

$$\mathbb{E}[R_i] - R_f = \beta_i(\mathbb{E}[R_m] - R_f), \tag{25}$$

where R_m represents the return of the market portfolio and $\mathbb{E}[R_m]$ is the expected return. $\mathbb{E}[R_m] - R_f$ is therefore the risk premium of the market portfolio, known as the market factor. In this case, β_i is a measure of an asset's sensitivity to market movements. It represents the **systematic risk** of an asset relative to the overall market. Systematic risk, also known as market risk, is the risk inherent to the entire market or market segment, which cannot be eliminated through diversification.

CAPM is a statement that the market portfolio is on the mean-variance frontier.

CAPM is intuitively appealing because it is built on a set of straightforward assumptions that lead to a clear understanding of market equilibrium. First, the model assumes that investors are rational and seek to maximize their utility. They do this by diversifying their portfolios to minimize risk, as diversification reduces the impact of any one asset's poor performance. This behavior ensures that only systematic risk remains relevant, which is risk inherent to the entire market. Second, the CAPM assumes that investors can borrow and lend at a risk-free rate and have homogeneous expectations, meaning they all have access to the same information and agree

on the expected returns and risks of securities. This creates a level playing field where no single investor has an advantage over others. In this context, the market reaches an equilibrium where the prices of securities fully reflect all available information. The equilibrium condition implies that the supply and demand for each asset are balanced, leading to the market portfolio being mean-variance efficient. This means the market portfolio, which includes all investable assets weighted by their market value, offers the highest expected return for a given level of risk.

A concept related to CAPM is the Security market line (SML). It displays the expected return of an individual asset as a function of beta. In equilibrium, the relationship between risk and return is linear, and the expected return on any asset is proportional to its beta. This relationship is encapsulated in SML.

3 Stochastic Discount Factor

3.1 Concept

The no-arbitrage principle is a foundational concept in finance, which asserts that it is impossible to construct a portfolio that yields a risk-free profit without any investment. If such opportunities existed, arbitrageurs would exploit them until the prices adjusted to eliminate the arbitrage. This principle leads us to the fundamental theorem of asset pricing, which states that in a market free of arbitrage, there exists at least one probability measure under which the discounted price processes of all tradable assets are martingales. This measure is often referred to as the risk-neutral measure or equivalent martingale measure.

To understand asset prices in a no-arbitrage framework, we introduce the concept of the Stochastic Discount Factor (SDF), also known as the *pricing kernel*. The SDF is a random variable that adjusts future cash flows to their present value, reflecting the time value of money and risk preferences of investors.

In a no-arbitrage framework, the present value of an asset can be expressed as:

$$P_t = \mathbb{E}[m_{t+1} \cdot X_{t+1}], \tag{26}$$

where P_t is the current price of the asset at time t, m_{t+1} is the stochastic discount factor between t and $t+1$, and X_{t+1} is the future payoff of the asset at time $t+1$. The SDF m_{t+1} essentially captures the idea that investors require compensation for the time value of money and for bearing risk. It adjusts future payoffs to present values in such a way that all assets, when priced appropriately, offer no arbitrage opportunities.

Now, consider the risk-free asset with a certain payoff R_f. If we invest one dollar today in this asset, we will get $1 + R_f$ at the end of next period. This means that $1 = \mathbb{E}[m_{t+1}(1 + R_f)]$, or:

$$\mathbb{E}[m_{t+1}] = \frac{1}{1 + R_f}. \tag{27}$$

For the excess return R^e_{it}. Since is zero-payment today, its price must be zero and therefore:

$$0 = \mathbb{E}[m_{t+1} R^e_{it}]. \tag{28}$$

We will make use of this equation later.

3.2 Mean-Variance Frontier and SDF

We have talked about mean-variance frontier, and we have talked about SDF. Asset pricing theory tells us that they are the same thing (Cochrane 2005):

There is a discount factor of the form $m = a + bR_{\mathbf{mv}}$ if and only if $R_{\mathbf{mv}}$ is on the mean-variance frontier, and $R_{\mathbf{mv}}$ is not the risk-free rate.

We explain what it means here. Start with $0 = \mathbb{E}[mR^e]$ (recall Section 3!), where m is the SDF and R^e is the excess return of some asset (I neglect subscripts for simplicity):

$$\begin{aligned} 0 &= \mathbb{E}[mR^e] \\ &= \mathbb{E}[m]\mathbb{E}[R^e] + \text{cov}(m, R^e) \\ &= \mathbb{E}[m]\mathbb{E}[R^e] + \rho(m, R^e)\sigma(m)\sigma(R^e). \end{aligned} \tag{29}$$

Therefore:

$$\frac{\mathbb{E}[R^e]}{\sigma(R^e)} = -\frac{\rho(m, R^e)\sigma(m)}{\mathbb{E}[m]}. \tag{30}$$

Since $-1 \leq \rho(m, R^e) \leq 1$, then we can derive the following relationship about the excess return and SDF, and it is known as Hansen and Jagannathan (1991) bound:

$$\frac{|\mathbb{E}[R^e]|}{\sigma(R^e)} \leq \frac{\sigma(m)}{\mathbb{E}[m]}. \tag{31}$$

Let's focus on the assets with a positive $\mathbb{E}[R^e]$, then it becomes:

$$\frac{\mathbb{E}[R^e]}{\sigma(R^e)} \leq \frac{\sigma(m)}{\mathbb{E}[m]}. \tag{32}$$

Note that the LHS above is just the definition of the Sharpe ratio, so the inequality means that the Sharpe Ratio of the investment portfolio has an upper limit. Observing this formula, we focus on two questions: (1) When does the equality hold? (2) What does the equality mean? The answer to the first question is simple: when the correlation coefficient $\rho(m, R^e) = -1$, the equality holds. For the second question, **equality implies that the asset is on the mean-variance frontier** (because these assets have the highest Sharpe Ratios), and the correlation coefficient between these assets and the SDF is exactly -1. This means that **all assets on the mean-variance frontier are**

perfectly negatively correlated with the SDF, and therefore they are all perfectly positively correlated with each other. Such a perfectly correlation between these assets and SDF implies that for any asset R_{mv} on the mean-variance frontier, there is a corresponding SDF representation:

$$m = a + bR_{\text{mv}}, \tag{33}$$

where a and b are coefficients.

4 Multi-Factor Models

We have introduced the single factor model:

$$\mathbb{E}[R_i^e] = \beta_i \mathbb{E}[R_{\text{mv}}^e]. \tag{34}$$

The multi-factor model is a natural extension to it by considering multiple factors, instead of a single factor:

$$\mathbb{E}[R_i^e] = \boldsymbol{\beta}_i^\top \boldsymbol{\lambda}, \tag{35}$$

where $K \times 1$ vector $\boldsymbol{\beta}_i$ represents factor exposures of asset i, and $K \times 1$ vector $\boldsymbol{\lambda}_i$ represents the risk premia of these factors. Given the equivalence between the mean-variance frontier and the factor model, **a multi-factor model is just a statement that a combination of the factors is on the mean-variance frontier.**

We can motivate multi-factor models from either Intertemporal Capital Asset Pricing Model (ICAPM) of Merton (1973) or the Arbitrage Pricing Theory (APT) of Ross (1976), where ICAPM generalizes SDF's expression to a linear combination of multiple factors, while APT starts from a statistical characterization that there are several common components or sources in asset returns, and therefore it suggests that one can start with a statistical analysis of the covariance matrix of returns to characterize common movement. Since we will later talk about latent factor models (which will be estimated from the covariance matrix of assets' returns), let me start with the APT view.

According to APT, the data generating process of assets' excess returns is governed by

$$\boldsymbol{R}_t^e = \mathbb{E}[\boldsymbol{R}^e] + \boldsymbol{\beta}\boldsymbol{v}_t + \boldsymbol{e}_t, \tag{36}$$

where $\boldsymbol{R}_t^e$ is an $N \times 1$ vector of assets' excess returns, $\boldsymbol{\beta}$ is an $N \times K$ matrix of factor exposures, $\boldsymbol{v}_t$ is a $K \times 1$ vector of zero-mean factor innovations defined as $\boldsymbol{v}_t = \boldsymbol{f}_t - \mathbb{E}[\boldsymbol{f}_t]$ with $\boldsymbol{f}_t$ being factor realization, and $\boldsymbol{e}_t$ is an $N \times 1$ vector of zero-mean idiosyncratic errors, such that $\mathbb{E}[\boldsymbol{e}_t|\boldsymbol{v}_t] = 0$.

When factors are tradable portfolios, it can be shown that factors' risk premia $\boldsymbol{\lambda} = \mathbb{E}[\boldsymbol{f}_t]$. To

see this, the no-arbitrage condition implies the existence of an SDF m of the form:

$$m = a - \boldsymbol{b}^\top(\boldsymbol{f}_t - \mathbb{E}[\boldsymbol{f}_t]), \tag{37}$$

satisfying the fundamental equation $\mathbb{E}[mR_i^e] = 0$. Without loss of generality, we can normalize m such that $\mathbb{E}[m] = 1$, which means that a takes the value of 1, and therefore m becomes $m = 1 - \boldsymbol{b}^\top(\boldsymbol{f}_t - \mathbb{E}[\mathbf{f}_t])$. To derive the linear factor model, start with $\mathbb{E}[mR_{it}^e] = 0$ and write

$$\begin{aligned} 0 \;=\; \mathbb{E}[mR_i^e] \;&=\; \mathbb{E}[m]\mathbb{E}[R_i^e] - \mathrm{cov}(R_{it}^e, \boldsymbol{f}_t^\top)\boldsymbol{b} \\ &=\; \mathbb{E}[R_i^e] - \mathrm{cov}(R_{it}^e, \boldsymbol{f}_t^\top)\boldsymbol{b}. \end{aligned} \tag{38}$$

Therefore, we can write

$$\mathbb{E}[R_i^e] = \mathrm{cov}(R_i^e, \boldsymbol{f}_t^\top)\boldsymbol{b}. \tag{39}$$

Next, plugging in the definition of $\boldsymbol{\beta}_i^\top \equiv \mathrm{cov}(R_i^e, \boldsymbol{f}_t^\top)\mathrm{var}(\boldsymbol{f}_t)^{-1}$ gives

$$\begin{aligned} \mathbb{E}[R_i^e] \;&=\; \mathrm{cov}(R_i^e, \boldsymbol{f}_t^\top)\mathrm{var}(\boldsymbol{f}_t)^{-1}\mathrm{var}(\boldsymbol{f}_t)\boldsymbol{b} \\ &=\; \boldsymbol{\beta}_i^\top \mathrm{var}(\boldsymbol{f}_t)\boldsymbol{b}, \end{aligned} \tag{40}$$

where $\boldsymbol{\lambda} \equiv \mathrm{var}(\boldsymbol{f}_t)\boldsymbol{b}$ are the risk premia of the factors. To see the meaning of $\boldsymbol{\lambda}$, multiplying $\boldsymbol{f}_t^\top$ on both sides of $m = 1 - \boldsymbol{b}^\top(\boldsymbol{f}_t - \mathbb{E}[\boldsymbol{f}_t])$ and then taking expectation yield,

$$\begin{aligned} \mathbb{E}[m\boldsymbol{f}_t^\top] \;&=\; \mathbb{E}[\boldsymbol{f}_t^\top] - \boldsymbol{b}^\top\mathbb{E}[(\boldsymbol{f}_t - \mathbb{E}[\boldsymbol{f}_t])\boldsymbol{f}_t^\top] \\ &=\; \mathbb{E}[\boldsymbol{f}_t^\top] - \boldsymbol{b}^\top\mathrm{var}(\boldsymbol{f}_t), \end{aligned} \tag{41}$$

or,

$$\mathbb{E}[m\boldsymbol{f}_t] = \mathbb{E}[\boldsymbol{f}_t] - \mathrm{var}(\boldsymbol{f}_t)\boldsymbol{b}. \tag{42}$$

Since factors are tradable portfolios and therefore $\boldsymbol{f}_t$ represents their excess returns, it follows that $\mathbb{E}[m\boldsymbol{f}_t] = 0$, and $\mathbb{E}[\boldsymbol{f}_t] = \mathrm{var}(\boldsymbol{f}_t)\boldsymbol{b} \equiv \boldsymbol{\lambda}$, or $\boldsymbol{\lambda} = \mathbb{E}[\boldsymbol{f}_t]$, i.e., the risk premium of a tradable factor is the expected value of its excess return, and $\boldsymbol{f}_t = \boldsymbol{\lambda} + \boldsymbol{v}_t$. Then, the multi-factor model follows:

$$\mathbb{E}[R_i^e] = \boldsymbol{\beta}_i^\top\boldsymbol{\lambda}. \tag{43}$$

This equation indicates that the expected excess return of a given asset is determined by its exposure to a set of factors and the risk premia of those factors, and it explains why assets have different expected returns in the *cross-section*. **Also, note that in the derivation above, we start from SDF and derive the beta factor model with exactly the same factors, and therefore it also shows that these two views are equivalent.** John Cochrane described this equivalence as:

An expected return beta model is equivalent to a discount factor that is a linear function of the factors in the beta model. This is an important and central result. It gives the connection between the discount factor formulation and the expected return-beta factor model formulation common in empirical work.

In practice, we usually add a term α_i, representing **pricing error**, to the model to match the empirical data:

$$\mathbb{E}[R_i^e] = \alpha_i + \boldsymbol{\beta}_i^\top \boldsymbol{\lambda}. \tag{44}$$

to account for several important aspects that the standard beta factor model might overlook. First, in practice, financial markets are highly complex, and a factor model may miss crucial factors that influence asset returns. Therefore, model misspecification is one key reason for including α_i. When the model omits important factors, α_i reflects the systematic mispricing resulting from this omission. This term helps to ensure that the model accounts for all relevant variables affecting asset returns, providing a more comprehensive understanding of the financial markets. Furthermore, even if the model includes all relevant factors, the estimation of these factors and their corresponding betas is based on historical data, which only represents a sample of the true population. This introduces sampling error and potential biases in the estimates. Statistical tests on α_i can indicate whether the deviations are due to random sampling error or if they signify more substantial model misspecification. Market inefficiencies and mispricing also justify the inclusion of α_i. Real-world financial markets are not always efficient, and various anomalies, behavioral biases, and market frictions can cause asset prices to deviate from their theoretical values. The inclusion of α_i allows the model to capture these inefficiencies, acknowledging that some portion of the asset's return could be attributed to market imperfections rather than purely systematic risk factors.

This cross-sectional equation (44) is both the heart and soul of empirical asset pricing and factor investing.

Besides, plugging $\mathbb{E}[R_i^e] = \alpha_i + \boldsymbol{\beta}_i^\top \boldsymbol{\lambda}$ into the data generating process and using the fact that $\boldsymbol{f}_t = \boldsymbol{\lambda} + \boldsymbol{v}_t$, we can re-write the data generating process of assets' excess returns as:

$$\mathbf{R}_t^e = \boldsymbol{\alpha} + \boldsymbol{\beta} \boldsymbol{f}_t + \boldsymbol{e}_t. \tag{45}$$

It can also be written for individual asset i's excess return:

$$R_{it}^e = \alpha_i + \sum_{k=1}^{K} \beta_{ik} f_{kt} + e_{it}, \tag{46}$$

where β_{ik} is the factor loading of asset i on factor k and f_{kt} is the factor k's excess return realization at time t. In general, the number of factors K is much smaller than the number of assets N. Of course, if there is no pricing error for asset i, then $\alpha_i = 0$.

5 Ad-Hoc Factor Models

While theory provides a framework for studying assets' returns under the multi-factor model framework, it does not offer much guidance on which factors should be included on the right-hand-side of the equation. Regarding the identification of factors, there are two main approaches and both have gained a lot of attention.

The first approach assumes either factors or factor exposures are known and observable, and academia and industry take different stances in this regard. Specifically, academia considers that factors are tradable portfolios that are constructed by using some firm characteristics, such as the HML and SMB factors of Fama and French (1993) (which are built by using the book-to-market ratio and market cap). In this case, factor realizations are the excess returns of those tradable portfolios, and factor loadings are determined by regressing assets' excess returns on factors. A related but different stance commonly adopted by industry practice (e.g. MSCI Barra model) assumes that factor loadings are observable. In this case, industry practice uses firm characteristics (usually after necessary normalization) directly as factor exposures, and factors' risk premia can be obtained by regressing assets' return on factor exposures. These two distinct perspectives also explain the different meanings regarding the term factors used by academia and industry. **For academia, the term factors means tradable portfolios. For industry, this term refers to firm characteristics and although not mentioned explicitly are indeed served as the factor exposures in the linear factor model framework.**

The second approach, as pioneered by Chamberlain and Rothschild (1983) and Connor and Korajczyk (1986), assumes both factors and factor exposures are latent. In this case, statistical techniques such as principal component analysis is employed to identify the factor model, and the factors are called *statistical factors*.

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